Dr. Travis J. Kessler

Easton, NY 12154
(978) 201-7710

travis.j.kessler@gmail.com
https://www.traviskessler.com
https://github.com/tjkessler

Summary: AI and machine learning engineer (PhD, Computer Engineering) with a proven record of impactful research, open-source software, and scalable AI systems for science, industry, and energy. Recognized for advancing computational methods and delivering practical solutions across multiple domains.

Leadership & Awards: Led and mentored technical teams and research projects, driving innovation and collaboration. Honored with the Dean's Gold Medal, Outstanding PhD Award, and multiple first-place finishes in engineering and innovation competitions.

Technical Expertise: Specialized in machine learning, deep learning, MLOps, and cloud architectures. Experienced in predictive modeling, simulation, and analytics. Published in top journals and conferences; creator of open-source tools for the research community.

Deployment Experience: Architected and deployed secure, production-grade AI platforms and web applications, integrating authentication, payments, and automated CI/CD pipelines. Delivered robust, scalable solutions for research, analytics, and commercial use.

TECHNICAL SKILLS

Programming & Tools
- Python, C/C++, MATLAB
- TensorFlow, PyTorch
- Scikit-learn, NumPy, Pandas
- Jupyter Notebook
- Ray, Dask

ML & Data Science
- Predictive/statistical modeling
- Regression, transformers
- LLM tuning, RAG, vector DBs
- Web scraping, data pipelines
- Data visualization

Cloud & DevOps
- AWS, GCP, Azure
- Docker, Git, SVN
- SQL (Postgres), NoSQL
- Cloud deployment, CI/CD
- Containerization, MLOps

EDUCATION

PhD, Computer Engineering
University of Massachusetts Lowell
May 2023

B.S., Computer Engineering
University of Massachusetts Lowell
May 2018

EXPERIENCE

Founding Technical Consultant *Apr. 2025 — Present*
Cognitive Chemistry Labs

- Engineered and deployed a highly scalable, serverless AI platform for computational chemistry and biology, accelerating high-throughput analysis and research workflows.
- Built and launched a full-stack web application (TypeScript frontend, FastAPI backend), integrating Supabase for authentication and database management, and Modal for dynamic, serverless compute resources.
- Delivered a user-centric website for account management, subscription billing, and compute job payments, integrating Stripe for secure, automated transactions.
- Instituted comprehensive unit testing and CI/CD pipelines, driving code quality and enabling seamless automated deployment to development and production environments.

Research Engineer *Oct. 2023 — Apr. 2025*
AIMdyn, Inc.

- Spearheaded development of data-driven solutions leveraging generative AI and deep learning to optimize forecasting, simulation, and analytic engine performance for strategic planning and discovery.
- Integrated scalable, reusable MLOps components (MLflow, CI/CD, containerization), expediting experimentation and deployment of machine learning models for simulation and COA validation.
- Orchestrated server cluster workflows and job scheduling systems, enabling high-throughput simulation and analytic engine operations for large-scale scenario exploration.
- Advanced the SCEPTER program by creating unscripted, goal-oriented agents that autonomously discover novel, relevant, and interpretable courses of action (COAs) in trusted military simulation environments.
- Devised and implemented algorithms to manage exponential growth of the global state-action space, enabling rapid exploration of large-scale military scenarios and achieving scenario exploration speeds up to 100,000x real-time.

Graduate Research Assistant *Jan. 2019 — Sep. 2023*
University of Massachusetts Lowell Energy & Combustion Research Laboratory

- Applied advanced machine learning and data engineering techniques for robust data preprocessing, modeling, and validation in alternative energy research projects.
- Leveraged optimization algorithms (including biologically-inspired methods) to fine-tune hyper-parameters in supervised and unsupervised learning workflows, boosting predictive model accuracy.
- Automated data preprocessing pipelines in Python, enhancing consistency, reproducibility, and accuracy for downstream statistical evaluation.
- Mentored research teams and provided technical leadership on software development, ensuring quality through code reviews, collaborative troubleshooting, and version control best practices (Git).

Implementation/DevOps Engineer *June 2018 — Jan. 2019*
Valora Technologies

- Engineered automated extraction workflows to convert complex legal, financial, and government documents into actionable data insights, improving client deliverables.
- Designed and optimized robust ETL pipelines, streamlining client document processing from intake to final data delivery and reducing turnaround time.
- Led onboarding and training for new Implementation/DevOps engineers, empowering teams to master advanced data mining, pipeline configuration, and DevOps best practices.

Undergraduate Research Assistant *June 2015 — May 2018*
University of Massachusetts Lowell Energy & Combustion Research Laboratory

- Developed advanced predictive models to accurately forecast chemical properties from molecular structures, driving innovation in QSAR/QSPR research.
- Designed and deployed high-performance neural network architectures for complex, multidimensional datasets, increasing model accuracy and reliability.
- Created and released open-source software packages for machine learning, feature extraction, and hyper-parameter optimization, enabling the research community with robust, reusable tools.
- Published impactful research in leading scientific journals and conferences, including *Fuel*, *The Journal of Open Source Software*, and the *ASME Internal Combustion Engine Fall Conference*.

SELECTED AWARDS & HONORS

Dean's Gold Medal for Outstanding Academic Achievement	*May 2023*
Computer Engineering Department Award for Outstanding Ph.D.	*May 2023*
Best Presentation, ASME ICEF 2019 Conference	*Oct. 2019*
1st place, Symbotic Warehouse Robot Prototyping Competition	*May 2018*
Innovative Technology Solution, UMass Lowell DifferenceMaker 50K Idea Challenge	*April 2017*
1st place, Francis College of Engineering Prototyping Competition	*Dec. 2016*

PUBLISHED WORKS

PB12. Fnu Gorky, Apolo Nambo, **Travis J. Kessler**, J. Hunter Mack, Maria L. Carreon. "CO_2 and HPDE Upcycling: A Plasma Catalysis Alternative". *Industrial & Engineering Chemistry Research* (2023). https://doi.org/10.1021/acs.iecr.3c02403

PB11. Amina SubLaban, **Travis Kessler**, Noah Van Dam, J. Hunter Mack. "Artificial Neural Network Models for Octane Number and Octane Sensitivity: A Quantitative Structure Property Relationship Approach to Fuel Design". *Journal of Energy Resources Technology* (2023). https://doi.org/10.1115/1.4062189

PB10. **Travis Kessler**, Amina SubLaban, J. Hunter Mack. "Predicting the Cetane Number, Sooting Tendency, and Energy Density of Terpene Fuel Additives". *ASME Internal Combustion Engine Division Fall Technical Conference* (2022). https://doi.org/10.1115/ICEF2022-91163

PB9. **Travis Kessler**, Thomas Schwartz, Hsi-Wu Wong, J. Hunter Mack. "Evaluating Diesel/Biofuel Blends Using Artificial Neural Networks and Linear/Nonlinear Equations". *ASME Internal Combustion Engine Division Fall Technical Conference* (2021). https://doi.org/10.1115/ICEF2021-67785

PB8. **Travis Kessler**, Thomas Schwartz, Hsi-Wu Wong, J. Hunter Mack. "Predicting the Cetane Number, Yield Sooting Index, Kinematic Viscosity, and Cloud Point for Catalytically Upgraded Pyrolysis Oil Using Artificial Neural Networks". *ASME Internal Combustion Engine Division Fall Technical Conference* (2020). https://doi.org/10.1115/ICEF2020-2978

PB7. **Travis Kessler**, Peter C. St. John, Junqing Zhu, Charles S. McEnally, Lisa D. Pfefferle, J. Hunter Mack. "A comparison of computational models for predicting yield sooting index". *Proceedings of the Combustion Institute* (2020). https://doi.org/10.1016/j.proci.2020.07.009

PB6. **Travis Kessler**, Thomas Schwartz, Hsi-Wu Wong, J. Hunter Mack. "Screening Compounds for Fast Pyrolysis and Catalytic Biofuel Upgrading Using Artificial Neural Networks". *ASME Internal Combustion Engine Division Fall Technical Conference* (2019). https://doi.org/10.1115/ICEF2019-7170

PB5. Sanskriti Sharma, Hernan Gelaf-Romer, **Travis Kessler**, J. Hunter Mack. "ECabc: A feature tuning program focused on Artificial Neural Network hyperparameters". *Journal of Open Source Software* (2019). https://doi.org/10.21105/joss.01420

PB4. **Travis Kessler**, Eric Sacia, Alexis Bell, J. Hunter Mack. "Artificial neural network based predictions of cetane number for furanic biofuel additives". *Fuel*, 206, 171-179 (2017). https://doi.org/10.1016/j.fuel.2017.06.015

PB3. **Travis Kessler**, Gregory Dorian, J. Hunter Mack. "Application of a Rectified Linear Unit (ReLU) Based Artificial Neural Network to Cetane Number Predictions". *ASME Internal Combustion Engine Division Fall Technical Conference* (2017). https://doi.org/10.1115/icef2017-3614

PB2. **Travis Kessler**, J. Hunter Mack. "ECNet: Large scale machine learning projects for fuel property prediction". *Journal of Open Source Software* (2017). https://doi.org/10.21105/joss.00401

PB1. **Travis Kessler**, Eric Sacia, Alexis Bell, J. Hunter Mack. "Predicting the Cetane Number of Furanic Biofuel Candidates Using an Improved Artificial Neural Network Based on Molecular Structure". *ASME Internal Combustion Engine Division Fall Technical Conference* (2016). https://doi.org/10.1115/icef2016-9383